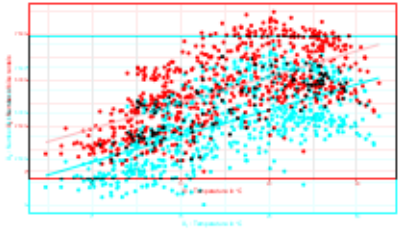
# GENERALIZED ADDITIVE MODEL (GAM)

- Hastie and Tibshirani (1986)

**Problem**: LM not great if features act on outcome non-linearly target variable is not linear



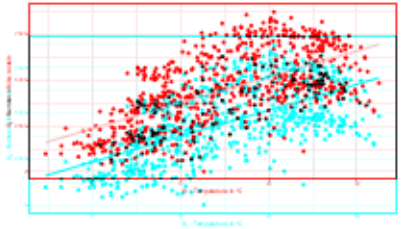
# GENERALIZED ADDITIVE MODEL (GAM)

▸ Hastie and Tibshirani (1986)

**Problem**: LM not great if features act on outcome non-linearly / relationship between features and target variable is not linear

**Workaround in LMs / GLMs:**

- Feature transformations (e.g., exp or log)
- Including high-order effects
- Categorization of features (i.e., intervals/ buckets of feature values)



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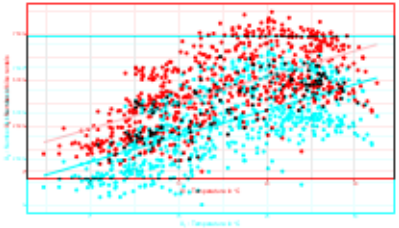


**Idea of GAMs:**

- Instead of linear terms $\theta_j x_j$, use flexible functions $f_j(x_j)$ $\rightsquigarrow$ splines

$$g(\mathbb{E}(y \mid \mathbf{x})) = \theta_0 + f_1(x_1) + f_2(x_2) + \ldots + f_p(x_p)$$

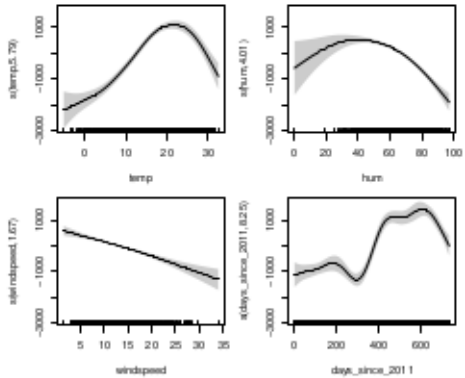
- Preserves additive structure and allows to model non-linear effects
- Splines have a smoothness parameter to control flexibility (prevent overfitting)
  $\rightsquigarrow$ Needs to be chosen, e.g., via cross-validation

# GENERALIZED ADDITIVE MODEL (GAM) - EXAMPLE

Fit a GAM with smooth splines for four numeric features of bike rental data

⇝ more flexible and better model fit but less interpretable than LM

| | edf | p-value |
|---|---|---|
| s(temp) | 5.8 | 0.00 |
| s(hum) | 4.0 | 0.00 |
| s(windspeed) | 1.7 | 0.00 |
| s(days_since_2011) | 8.3 | 0.00 |

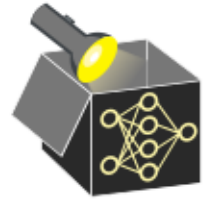



**Interpretation**

- Interpretation needs to be done visually and relative to average prediction, also see PDPs
- Edf (effective degrees of freedom) represents complexity of smoothness

# MODEL-BASED BOOSTING

Bühlmann and Yu 2003



- Boosting iteratively combines weak base learners to create powerful ensemble
- Idea: Use simple BLs (e.g univariate, with splines) to ensure interpretability
- Possible to combine BL of same type (with distinct parameters $\boldsymbol{\theta}$ and $\boldsymbol{\theta}^*$):

$$b^{[j]}(\mathbf{x}, \boldsymbol{\theta}) + b^{[j]}(\mathbf{x}, \boldsymbol{\theta}^*) = b^{[j]}(\mathbf{x}, \boldsymbol{\theta} + \boldsymbol{\theta}^*)$$

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$$b^{[j]}(\mathbf{x}, \theta) + b^{[j]}(\mathbf{x}, \theta^*) \equiv b^{[j]}(\mathbf{x}, \theta + \theta^*)$$

- In each iteration, fit a set of BLs, add best one to model (with step-size $\nu$):

$$\begin{aligned}
\hat{f}^{[1]} &= \hat{f}_0 + \nu b^{[3]}(\mathbf{x}_3, \theta^{[1]}) \\
\hat{f}^{[2]} &= \hat{f}^{[1]} + \nu b^{[3]}(\mathbf{x}_3, \theta^{[2]}) \\
\hat{f}^{[3]} &= \hat{f}^{[2]} + \nu b^{[1]}(\mathbf{x}_1, \theta^{[3]}) \\
&= \hat{f}_0 + \nu \left( b^{[3]}(\mathbf{x}_3, \theta^{[1]} + \theta^{[2]}) + b^{[1]}(\mathbf{x}_1, \theta^{[3]}) \right) \\
&= \hat{f}_0 + \hat{f}_3(\mathbf{x}_3) + \hat{f}_1(\mathbf{x}_1)
\end{aligned}$$

- Final model is additive GAM, we can read off effect curves

# MODEL-BASED BOOSTING - LINEAR EXAMPLE

Simple case: Use linear model with single feature (including intercept) as BL

- Idea: Use simple linear BL to ensure interpretability (in general also spline BL possible)

$$b^{[j]}(x, \theta) = x_j \theta + \theta_0 \quad \text{for } j = 1, \dots, p \quad \rightsquigarrow \text{ordinary linear regression}$$

- Possible to combine linear BL of same type (with distinct parameters $\theta$ and $\theta^*$):

$$b^{[j]}(\mathbf{x}, \theta) + b^{[j]}(\mathbf{x}, \theta^*) = b^{[j]}(\mathbf{x}, \theta + \theta^*)$$

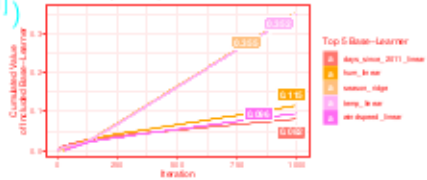
- Here: Interpretation of weights as in LM
- After many iterations, it converges to same solution as LM
- In each iteration, fit a set of BLs and add the best BL to previous model (using step-size $\nu$):

$$\hat{f}^{[m]} = \hat{f}^{[m-1]} + \nu b^{[3]}(\mathbf{x}_3, \theta^{[1]})$$

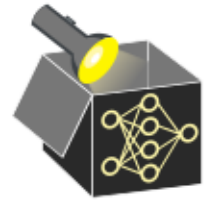
| 1000 iter. with $\nu = 0.1$ | Intercept | Weights |
|---|---|---|
| days_since_2011 | -1791.06 | 4.9 |
| hum | 1953.05 | -31.1 |
| season | 0 | WINTER: -323.4<br>SPRING: 539.5<br>SUMMER: -280.2<br>FALL: 67.2 |
| temp | -1839.85 | 120.4 |
| windspeed | 725.70 | -56.9 |
| offset | 4504.35 | |

$\Rightarrow$ Converges to solution of LM

Relative frequency of selected BLs across iterations



# MODEL-BASED BOOSTING - LINEAR EXAMPLE



Simple case: Use linear model with single feature (including intercept) as BL

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$$b^{[j]}(x, \theta) = x_j\theta + \theta_0 \quad \text{for } j = 1, \dots p \quad \rightsquigarrow \text{ordinary linear regression}$$

- Possible to combine linear BL of same type (with distinct parameters $\theta$ and $\theta^*$):

$$b^{[j]}(\mathbf{x}, \theta) + b^{[j]}(\mathbf{x}, \theta^*) = b^{[j]}(\mathbf{x}, \theta + \theta^*)$$

- Here: Interpretation of weights as in LM
- After many iterations, it converges to same solution as LM
- Early stopping allows feature selection & may prevent overfitting (regularization)

$$\hat{f}^{[1]} = \hat{f}^{[0]} + \nu b^{[3]}(\mathbf{x}_3, \theta^{[1]})$$

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| offset | 4504.35 | |

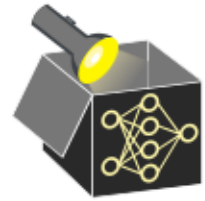
$\Rightarrow$ Converges to solution of LM

| 20 iter. with $\nu = 0.1$ | Intercept | Weights |
|---|---|---|
| days_since_2011 | -1210.27 | 3.3 |
| season | 0 | WINTER: -276.9<br>SPRING: 137.6<br>SUMMER: 112.8<br>FALL: 20.3 |
| temp | -1118.94 | 73.2 |
| offset | 4504.35 | |

$\Rightarrow$ 3 BLs selected after 20 iter. (feature selection)

# LINEAR EXAMPLE: INTERPRETATION





- Recall: Boosting iteratively combines weak base learners (BL)
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- In each iteration, fit a set of BLs and add the best BL to previous model (using step-size $\nu$):

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$$= \hat{f}_0 + \hat{f}_3(\mathbf{x}_3) +$$

**Feature importance:** aggregated change in risk in each iteration per feature

- E.g. iteration 1: $\text{days_since_2011}$ with risk reduction (MSE) of 140,782.94
- For every iteration the change in risk can be attributed to a feature
- Possible to combine linear BL of same type (with distinct parameters $\theta$ and $\theta^*$):

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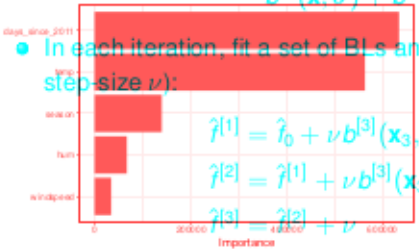
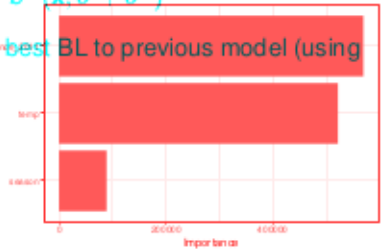
1000 iterations




In-bag-risk: 434,686.0
OOB risk (10-fold CV): 446,450.0

20 iterations




In-bag-risk: 693,505.0
OOB risk (10-fold CV): 705,776.0

$\Rightarrow$ Difference in risk: 258,819.0
Difference in OOB risk: 259,326.0

# NON-LINEAR EXAMPLE: INTERPRETATION

- Fit model on bike data with different BL types (1000 iter.) Daniel Schalk et al. 2018
- BLs: linear and centered splines for numeric features, categorical for season
- Idea: Use simple linear BL to ensure interpretability (in general also spline BL possible)
- Possible to combine linear BL of same type (with distinct parameters $\theta$ and $\theta^*$):

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\end{aligned}$$

- Final model is additive (as GAMs), where each component function is interpretable

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- Fit model on bike data with different BL types (1000 iter.)

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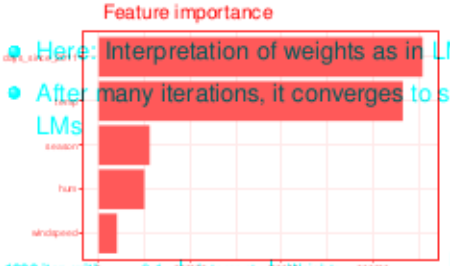

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- After many iterations, it converges to same solution as least square estimate of LMs

1000 iter. with $\nu = 0.1$

| | Intercept | Weights |
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| days_since_2011 | -1791.06 | 4.9 |
| season | 0 | SPRING: 539.5<br>SUMMER: 989.2<br>FALL: 87.2 |
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Relating frequency of selected BLs across iterations

⇒ In-bag-risk: 250,202.0 ; OOB risk (10-fold CV): 267,497.0 (difference to lin. example: 178,953.0)

⇒ In-bag-risk: 434,686.0 ; OOB risk (10-fold CV): 446,450.0 (previous lin. example with 1000 iter.)

- Feature importance (risk reduction over iter.)

  $\rightsquigarrow$ days_since_2011 most important

- Total effect for days_since_2011

  $\rightsquigarrow$ Combination of partial effects of linear BL and centered spline BL

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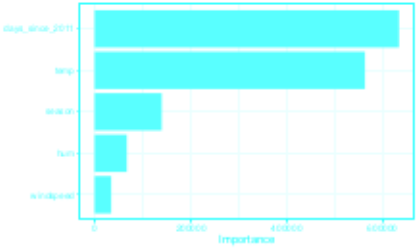


Overall risk: 434,686.0
OOB risk (10-fold CV): 446,450.0

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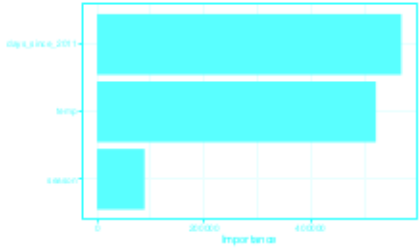


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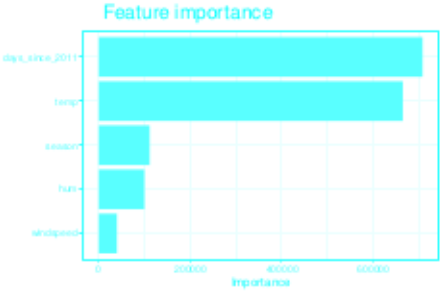


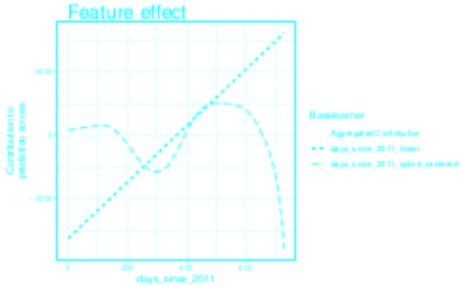


⇒ In-bag-Risk: 250,202.0 ; OOB risk (10-fold CV): 267,497.0 (Difference: 178,953.0)

- Feature importance (risk reduction over iter.) ⇝ `days_since_2011` most important
- Total effect for `days_since_2011`
  ⇝ Combination of partial effects of linear BL and centered spline BL